1. Introduction

Language model performance depends on architectural design and input embedding choices, which can influence predictive quality, computational efficiency, and text generation. This study compares different model architectures and embedding configurations to examine their effects on next-token prediction and downstream classification performance. It further investigates whether language-model predictive quality aligns with the effectiveness of learned representations in downstream tasks.

2. Methodology

2.1 Data, Preprocessing and Environment

WikiText-2 Raw [1], consisting of Wikipedia articles, was used for language-model training and evaluation with its official train, validation, and test splits. Three downstream datasets were selected: IMDb movie reviews [2] for binary sentiment classification, AG News articles [3] for four-class topic classification, and Wikipedia-derived DBpedia-14 descriptions [3, 7] for 14-class entity classification. For each downstream task, stratified subsets of 4,000 training, 1,000 validation, and 2,000 test examples were constructed, with the official test split reserved for final evaluation. Since DBpedia and WikiText-2 share a Wikipedia origin, potential article overlaps were screened through title matching, and 181 training and 29 test candidates were excluded before sampling.

Text was lowercased and tokenized using whitespace, with WikiText-specific punctuation markers normalized. Article boundaries and structural information were preserved using special tokens, while standalone integers occurring fewer than 20 times in WikiText-2 training data were mapped to <num>. A shared 10,000-token vocabulary was constructed exclusively from the WikiText-2 training split, covering 92.14% of ordinary training tokens; out-of-vocabulary tokens were mapped to <unk>. The same vocabulary and numerical normalization rules were applied to downstream texts, with additional HTML cleanup for IMDb.

Experiments were conducted on a Windows 10 laptop with an AMD Ryzen 7 PRO 4750U CPU and 16 GB RAM, using Python 3.8.10 and PyTorch 2.0.1 (CPU).

2.2 Language Models

Four next-token language models were compared: an interpolated 4-gram model with add-k smoothing, a vanilla RNN, an LSTM [4], and a causal Transformer [5]. The three neural models used randomly initialized, trainable 100-dimensional input embeddings. RNN and LSTM each contained one recurrent layer with 128 hidden units, while the Transformer used two layers, four attention heads, and a feed-forward dimension of 256.

The neural models were trained on article-bounded windows containing up to 64 prediction targets, with one-token overlap at window boundaries. Loss masking ensured that each target was counted once, and recurrent states were reset between windows. All three used Adam (learning rate 0.001), batch size 32, gradient clipping at 1.0, random seed 42, and 15 training epochs, with eight CPU threads. The interpolated 4-gram model used add-k smoothing with k = 10^-4 and interpolation weights (0.20, 0.45, 0.30, 0.05) for orders one through four, selected using validation perplexity.

2.3 Embedding Variants

Using the same Transformer architecture and training protocol, three input-embedding settings were compared: randomly initialized trainable embeddings, self-trained frozen GloVe embeddings, and public pretrained frozen GloVe embeddings.

Self-trained GloVe vectors [6] were learned from the WikiText-2 training corpus using a context radius of five and 20 training epochs. Public embeddings were obtained from Stanford GloVe 6B [6] (100 dimensions), covering 9,907 of the 9,993 ordinary vocabulary entries (99.14%). The 86 unmatched public vectors were randomly initialized, while the entire input-embedding table was frozen in both GloVe variants. All remaining Transformer parameters were trained from scratch.

2.4 Downstream Representations

The five trained neural language models were frozen and used as feature extractors. Each downstream document was divided into independent chunks of up to 64 tokens without truncation. Token representations were extracted from the recurrent layer or final Transformer layer before vocabulary projection; causal masking was retained for Transformer models. Hidden states were averaged across all tokens, including the initial <bos> token, to produce a fixed-length representation; no padding was introduced during extraction.

For each representation, a standardized logistic-regression classifier was trained on the same dataset splits. The regularization parameter C{0.1,1,10} was selected using validation Macro-F1.

2.5 Evaluation Protocol

Language-model prediction quality was evaluated using average next-token cross-entropy and perplexity on the same 240,628 WikiText-2 test targets, excluding masked and padded positions. Neural models used independent 64-target windows; the 4-gram used article-continuous contexts of up to three preceding tokens. Computational cost was compared using cumulative training-and-validation runtime, serialized model size, and teacher-forced test throughput (target tokens/s). Neural evaluation timing included DataLoader iteration, forward passes, and loss calculation, but excluded model loading and corpus preprocessing.

Generated text was compared using matched prompts and decoding settings, including greedy decoding and temperature-0.8 sampling with top-k=40, both prohibiting generated <unk> tokens. Downstream classification was evaluated using held-out accuracy and Macro-F1, supplemented by paired error analysis and representative misclassifications.

3. Results

3.1 Language Model Prediction Quality and Computational Cost (Part I)

Table 1 compares the four language models in terms of held-out prediction quality, model size, and computational cost.

Table 1. Language-model prediction quality and computational cost.

Model	Model size	Checkpoint	Training time	Test PPL ↓	Eval. tokens/s ↑
4-gram	3.37M entries	53.72 MiB	12.49 s*	159.40	111,974
RNN	2.32M params	26.76 MiB	95.79 min	121.41	11,982
LSTM	2.41M params	27.77 MiB	95.65 min	117.55	11,755
Transformer	**2.21M params**	**25.50 MiB**	107.52 min	**105.84**	**12,309**

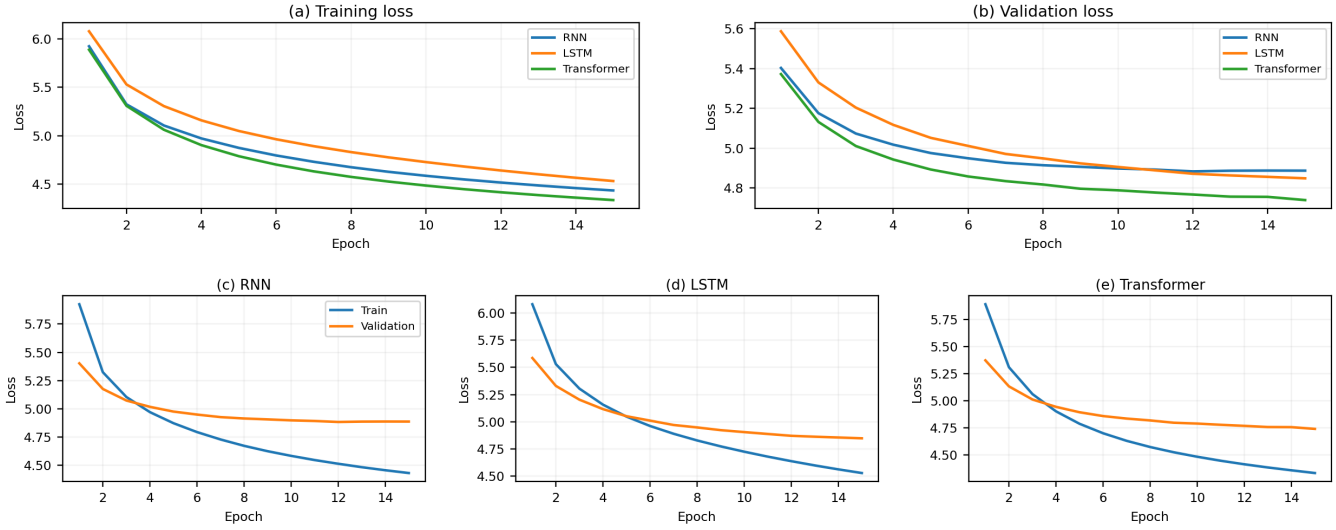
The 4-gram time covers count construction only, excluding preprocessing, saving, and hyperparameter search; neural-model times are cumulative training-and-validation wall-clock times across initial and resumed training, measured before each epoch's checkpoint save. Checkpoint sizes include serialized metadata. Throughput measures teacher-forced evaluation on the same test targets, not autoregressive generation speed.

The Transformer achieved the lowest test perplexity (105.84), followed by LSTM (117.55), RNN (121.41), and the 4-gram model (159.40). Despite having the fewest parameters among the neural models (2.21M), the Transformer required the longest training time (107.52 minutes), illustrating a trade-off between predictive quality and computational cost.

The 4-gram model achieved substantially higher evaluation throughput than the neural models, but its prediction quality remained lower after three rounds of validation-based hyperparameter search. The best validation perplexity improved from 185.55 in the first round to 172.47 and finally 170.32, with diminishing gains from further refinement. These results highlight the efficiency of the count-based baseline under the measured CPU workload, alongside the limitations of the selected smoothing and interpolation scheme.

3.2 Learning Behavior (Part I)

Figure 1. Training and validation loss trajectories of RNN, LSTM, and Transformer over 15 epochs. Panels (a–b) compare training and validation loss across models; panels (c–e) show both curves for each model.

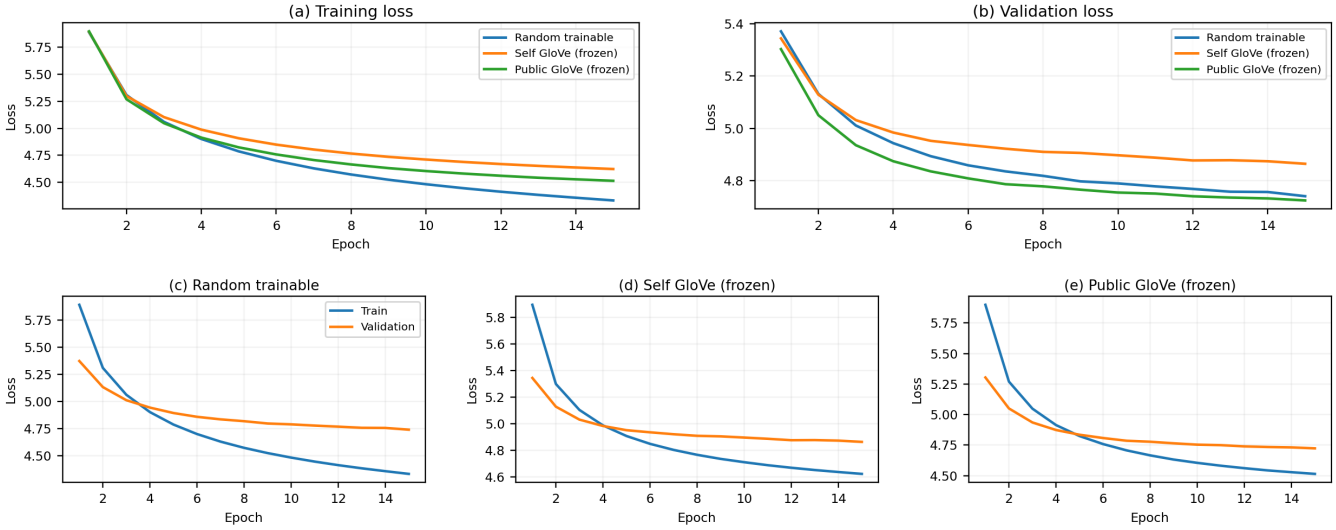


The Transformer maintained the lowest validation loss throughout training. Although RNN initially outperformed LSTM on validation, LSTM overtook RNN at epoch 11 and continued improving through epoch 15. RNN reached its best validation loss at epoch 12 (4.8838), followed by a slight increase, whereas LSTM and Transformer achieved their lowest recorded validation losses at epoch 15 (4.8489 and 4.7398, respectively). This change in ranking illustrates how conclusions about relative model performance can depend on the training budget.

Across all three models, training loss continued decreasing, while validation improvements became progressively smaller, widening the train–validation gap. By epoch 15, these gaps reached 0.4552 for RNN, 0.3190 for LSTM, and 0.4071 for Transformer. The divergence suggests increasing overfitting risk as training progresses, although the continued validation improvements of LSTM and Transformer indicate that additional training could still yield further generalization gains. Their eventual convergence behavior beyond 15 epochs remains undetermined.

3.3 Embedding Ablation (Part II)

Figure 2. Training and validation loss trajectories for the three Transformer embedding configurations over 15 epochs. Panels (a–b) compare training and validation losses across variants, while panels (c–e) show both curves for each variant.



Public GloVe achieved the lowest test perplexity (105.27), narrowly outperforming random trainable embeddings (105.84), while self-trained GloVe performed worse (119.86). The learning curves reveal a more interesting distinction: random trainable embeddings achieved the lowest final training loss (4.3327), whereas both frozen variants exhibited slower training-loss reduction and smaller train–validation gaps. Nevertheless, public GloVe achieved the best validation loss (4.7235),

suggesting that freezing sufficiently informative embeddings can constrain training-set fitting without sacrificing validation performance.

Self-trained GloVe substantially optimized its own objective, reducing weighted loss from 0.0583 to 0.0093 over 20 epochs, with 294.6 seconds of embedding optimization. Yet its frozen Transformer achieved the highest validation loss (4.8641). This contrast suggests that convergence on a word-embedding objective does not guarantee suitability for next-token prediction. The smaller WikiText-2 pretraining corpus and the inability to adapt frozen vectors may both contribute, whereas public GloVe benefits from substantially larger external training data.

Despite freezing approximately 45% of the Transformer parameters, training time was not substantially reduced. The random-trainable, self-GloVe, and public-GloVe variants required approximately 107.52, 113.47, and 101.34 minutes, respectively. Self-GloVe pretraining added only 294.6 seconds of optimization, a modest cost relative to Transformer training. All three embedding settings showed smooth recorded training and validation loss trajectories without obvious instability, despite different convergence rates.

All three variants reached their lowest recorded validation losses at epoch 15, although their longer-term convergence remains unknown. Random trainable embeddings showed a faster validation-loss reduction than public GloVe over the final three epochs, suggesting that their performance gap may change with additional training. We strongly hypothesize that initializing the Transformer with pretrained GloVe embeddings while keeping them trainable would outperform random initialization, particularly with public GloVe. Combining pretrained representations with adaptation to the language-modeling objective is therefore worth investigating in future work.

3.4 Text Generation (Parts I–II)

The 24 fixed prompts were divided into eight groups based on three dimensions: Wiki-like versus non-Wiki-like style, simple versus moderate vocabulary complexity, and short versus long context, with three prompts per group. All six models generated up to 50 tokens per prompt using greedy decoding and temperature-controlled top-k sampling. For the latter, temperature 0.8 and top-k 40 were selected through preliminary tuning; both methods prohibited generating <unk>.

Overall, the generated continuations frequently lacked grammatical fluency and sustained semantic coherence. Greedy decoding often produced repetitive loops, while stochastic sampling increased lexical diversity but did not resolve grammatical errors and sometimes introduced greater topic drift and unusual word choices.

Table 2. Generation diversity and repetition under two decoding strategies.

Model	Greedy Dist-2 ↑	Greedy Rep-3 ↓	Sampling Dist-2 ↑	Sampling Rep-3 ↓
4-gram	0.374	0.615	0.951	0.006
RNN	**0.580**	**0.346**	**0.956**	**0.003**
LSTM	0.462	0.483	0.948	0.006
Transformer	0.531	0.350	0.940	0.010
Self GloVe	0.461	0.408	0.915	0.012
Public GloVe	0.495	0.384	0.935	0.008

Sampling consistently increased Distinct-2 and reduced repeated 3-grams across all models. RNN achieved the best diversity and repetition scores under both decoding strategies despite having higher test perplexity than the Transformer variants. However, these metrics measure only lexical variation, and better scores do not necessarily indicate higher generation quality. Prompt-category comparisons revealed small, model-dependent differences without a consistent advantage for Wiki-like, simpler, or longer prompts.

For the shared prompt *“the river flows”*, greedy decoding produced repeated phrases such as *“missouri river river river”* (self-GloVe), while sampling largely eliminated these loops. Under sampling, the Transformer retained river-related geographical references, including Omaha and the Kansas River, although its syntax remained unnatural. Public GloVe instead shifted toward birds and vegetation despite achieving slightly lower test perplexity. These examples illustrate that lower perplexity does not

guarantee better topic consistency in individual generations.

Table 3. Representative continuations for the shared prompt “the river flows” (first 22 generated tokens; excerpts unedited).

Model	Greedy decoding	Temperature 0.8, top-k 40
Ngram	northeast to the city of the the grand slam tournament. the first time in the united states, and the the grand slam ...	northeast to the first division). for the first and, after the " rachel grew increasingly frustrated by the constant danger of the ...
Rnn	the road interchange (the north of the mississippi river), the largest number of the city 's main armament of the city. the ...	west (<num>) at the storm were to be the first ship to the northeast line while it now a larger area in ...
Lstm	the highway was a tropical storm, and the first two of the first time in the late 19th century, the storm was ...	west) the missouri river has made a single - up highway to the atlantic ocean united states with the 3rd light horse ...
Transformer	from the north sea, the north sea, missouri river, missouri river, missouri river, and the missouri river, and the missouri river river ...	through the city of omaha was a significant area of the city 's largest river avenue located near the confluence of kansas ...
Self Glove	the missouri river river river river, the missouri river river, the missouri river river river river river, and the missouri river river ...	into the city of zagreb river and northern australia, the south dakota was the fifth terrace over three rivers. these areas were ...
Public Glove	, the missouri river, missouri river, missouri river, and the platte river, and the missouri river. the missouri river was the missouri ...	is a first known as a " dry ". the fur is a metaphor used by a variety of staple birds in ...

An exploratory AI-assisted evaluation assessed generated continuations under both decoding strategies, scoring repetition, local fluency, topic consistency, and internal coherence on four equally weighted 0–5 scales. All six models received low and relatively similar overall scores, indicating generally weak generation quality without a clear model-level advantage. Since the evaluation relied on a single AI evaluator and abbreviated reviews of most continuations, the scores serve only as supplementary observations and cannot establish a reliable model ranking.

3.5 Downstream Representation Analysis (Part III)

Table 4. Language-model test perplexity and downstream classification accuracy (Acc) and Macro-F1.

Model	LM PPL ↓	IMDb Acc	IMDb F1	AG Acc	AG F1	DB Acc	DB F1
RNN	121.41	0.6845	0.6844	0.6125	0.6112	0.8035	0.8035
LSTM	117.55	0.6865	0.6865	0.6015	0.6010	0.7990	0.7983
Transformer	105.84	0.6730	0.6730	0.6715	0.6701	0.8510	0.8505
Self GloVe	119.86	0.7050	0.7050	0.6540	0.6526	0.8340	0.8342
Public GloVe	**105.27**	**0.7200**	**0.7199**	**0.8145**	**0.8142**	**0.9055**	**0.9053**
Uniform random*	—	0.500	≈0.500	0.250	≈0.250	0.071	≈0.071

Uniform-random reference values assume balanced test labels and uniform predictions; Macro-F1 values are large-sample approximations, not measured classifier results.

Public GloVe achieved the highest Macro-F1 across all three downstream tasks. Its advantage over the random-initialized Transformer was particularly pronounced on AG News (0.8142 versus 0.6701), despite only a small difference in language-model perplexity (105.27 versus 105.84). All five representation-based classifiers substantially exceeded the uniform-random reference levels across all three tasks. Model rankings also varied by task: the Transformer ranked second on AG News and DBpedia but last on IMDb, whereas self-GloVe ranked second on IMDb and third on both topic-related tasks.

Paired error analysis revealed that public GloVe corrected 297, 375, and 184 Transformer errors on IMDb, AG News, and DBpedia, respectively, while introducing 203, 89, and 75 new errors. Although public GloVe achieved net improvements across all three tasks, these regressions show that its predictions were not uniformly better on individual examples.

Representative IMDb cases illustrate both directions of this trade-off. For a negative review containing *“i have never fallen asleep whilst watching a movie before”* and *“avoid at all costs”*, the Transformer predicted positive sentiment, whereas public GloVe classified it correctly. Conversely, a positive review stating *“badly acted and directed, poorly scripted. who cares?”* was correctly classified by the Transformer but misclassified as negative by public GloVe. This example highlights how prominent negative expressions can conflict with a review's overall sentiment, and how public GloVe, despite its superior aggregate performance, can still perform worse on particular examples.

Finally, lower language-model perplexity did not consistently correspond to higher downstream Macro-F1. RNN and LSTM achieved very similar scores across all three tasks; despite LSTM's lower perplexity, its F1 was slightly below RNN's on two of them. Similarly, the random-initialized Transformer achieved substantially lower perplexity than self-GloVe (105.84 versus 119.86), yet self-GloVe performed better on IMDb (0.7050 versus 0.6730), while the gaps on AG News and DBpedia were comparatively small (0.0175 and 0.0163). These ranking reversals indicate that next-token prediction quality alone is insufficient to characterize the usefulness of frozen representations for classification.

4. Discussion and Limitations

The results demonstrate that next-token perplexity, generation behavior, and downstream classification performance capture complementary aspects of language-model capability. Lower perplexity does not consistently imply greater lexical diversity or higher downstream Macro-F1, yet perplexity remains an essential measure of predictive generalization on unseen text. Conversely, strong performance on a single downstream task does not establish overall model superiority. For task-specific applications, model selection should prioritize performance on the intended task, while broader language-model evaluation should combine perplexity, generation quality, and multiple diverse downstream benchmarks.

The embedding comparison also requires distinguishing pretrained initialization from embedding trainability. Because both GloVe variants were frozen while random embeddings remained trainable, the weaker performance of self-GloVe cannot be interpreted as evidence against pretrained embeddings in general. Allowing pretrained embeddings to adapt to the language-modeling objective could yield different outcomes, especially given the strong performance of frozen public GloVe.

The three downstream datasets were partly selected to explore whether similarity to WikiText-2 influenced representation transfer. However, after screening potential DBpedia article overlaps, unigram Jensen–Shannon divergence differed only modestly across datasets (0.320–0.337 bits), providing no clear separation in measured domain similarity. Since the tasks also differed in classification objectives and label counts, their Macro-F1 scores cannot isolate a domain-similarity effect. Future studies could use more closely controlled downstream tasks to investigate this relationship.

Several design choices limit the generality of these findings. Training was restricted to 15 epochs, before the eventual convergence or overfitting behavior of several models became apparent. Independent 64-token windows also prevented recurrent states from carrying information across segments, potentially limiting the benefits of LSTM memory mechanisms and contributing to the similar RNN/LSTM downstream results. Finally, matching neural models approximately by parameter count and training data enabled a controlled comparison at one scale, but architectures may respond differently to increased model size, data, or computation. Their rankings and performance gaps could therefore change under larger or longer training regimes.

5. Conclusion

This study compared language-model architectures, embedding configurations, and frozen representations across prediction, generation, and classification tasks. The Transformer achieved the strongest predictive performance among the original architectures, while public GloVe embeddings further improved perplexity and consistently achieved the highest downstream Macro-F1. However, model rankings varied across evaluation dimensions and tasks. These findings highlight the importance of selecting models according to their intended applications and evaluating general-purpose language models through complementary measures rather than a single performance metric.

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Appendices

Appendix A. Experimental Configurations and Training

Table A1. Main experimental settings.

Setting	RNN	LSTM	Transformer / GloVe variants
Input embedding	100-d, trainable	100-d, trainable	100-d; trainable (random) or frozen (GloVe)
Architecture	1 recurrent layer, hidden size 128	1 recurrent layer, hidden size 128	2 layers, 4 attention heads, FFN size 256
Training	Adam, LR 0.001; batch 32; 15 epochs	Same	Same
Sequence processing	Article-bounded 64-target windows	Same	Same; causal masking
Other settings	Seed 42; gradient clip 1.0; 8 CPU threads	Same	Same

The three neural architectures were matched approximately in total parameter count (2.21–2.41 million). The Part II comparison reused the random-initialized Transformer and trained two additional Transformer variants with frozen input embeddings. All models used the same 10,000-token WikiText-2-derived vocabulary.

Table A2. Selected n-gram configurations across validation searches.

Search stage	Best validation PPL	Configuration / observation
Initial search	185.55	Preliminary smoothing and interpolation search
Refinement	172.47	3-gram; add-k=10^{-4}; weights (0.20, 0.50, 0.30)
Final search	170.32	4-gram; add-k=10^{-4}; weights (0.20, 0.45, 0.30, 0.05)

Self-trained GloVe used a context radius of five and 20 epochs, processing 3,417,082 distinct co-occurrence pairs per epoch. Its weighted training loss decreased from 0.05832 to 0.00929. Optimization required 294.6 s; the recorded end-to-end GloVe run took 324.5 s, including preparation and co-occurrence construction. See results/glove/20261014_183352_778602/metrics.csv and summary.json. Full neural epoch histories are in the selected run directories under results/rnn/, results/lstm/, results/transformer/, and results/part2/.

Appendix B. Prompt Design and Generation Evaluation

Table B1. The 24 prompts in the eight crossed groups (three prompts per group).

Style	Complexity	Length	Prompts (IDs and text)
wiki like	simple	short	1: the city was; 2: the island has; 3: the river flows
non wiki like	simple	short	4: i feel happy; 5: we should go; 6: the air smells
wiki like	moderate	short	7: the historical district; 8: the regional economy; 9: the environmental impact
non wiki like	moderate	short	10: i noticed something; 11: the situation seems; 12: we discussed possible
wiki like	simple	long	13: the city developed around the river and became an important centre for trade during the nineteenth century; 14: the island is home to several species of birds and has a long history of human settlement; 15: the museum opened in the early twentieth century and contains a collection of paintings from different periods
non wiki like	simple	long	16: i went to the park after lunch and saw a group of children playing near the water; 17: we planned to visit the market in the morning but decided to stay home because it started raining; 18: i looked out of the window in singapore and wondered whether the sky would be clearer tomorrow
wiki like	moderate	long	19: the regional authorities introduced new environmental measures after researchers reported changes in air quality across several cities; 20: the historical building underwent extensive restoration before it reopened to the public as a cultural centre; 21: the university announced a collaborative research programme examining the relationship between urban development and public health
non wiki like	moderate	long	22: i checked the air quality information before leaving home because the sky looked unusually hazy that morning; 23: my friend suggested changing our outdoor plans if the haze became worse during the afternoon in singapore; 24: we were discussing whether the recent changes in visibility were related to weather conditions or regional smoke

Table B2. Exploratory AI-assisted evaluation rubric (0–5 per dimension).

Dimension	Score 0	Score 3	Score 5
Repetition	Severe mechanical loops	Some unnecessary repetition	No material repetitive looping
Local fluency	Largely unintelligible	Some locally natural phrases/ sentences	Mostly natural grammar and phrasing
Topic consistency	Unrelated to prompt	Partially maintains prompt topic	Sustains and develops prompt topic
Internal coherence	Disconnected content	Some coherent local progression	Coherent relationships throughout

Intermediate scores 1, 2, and 4 denote increasing quality between these anchors. Four dimensions were weighted equally; the aggregate is exploratory, not a validated human-evaluation benchmark. Most continuations received abbreviated single-pass AI review, so small score differences should not be interpreted as robust model rankings.

Table B3. Exploratory mean composite scores by model and decoding (0–5).

Model	Greedy	Temperature 0.8
4-gram	0.04	1.39
RNN	0.83	1.69
LSTM	0.51	1.57
Transformer	0.98	1.87
Self GloVe	0.75	1.72
Public GloVe	0.94	1.82

Table B4. Illustrative AI-assisted scores for selected continuations. Each excerpt is an uncorrected opening portion of the generated text. R = repetition, F = local fluency, T = topic consistency, C = internal coherence; all are scored from 0 to 5. The composite is their arithmetic mean.

Prompt	Model / decoding	Generated continuation (excerpt)	R	F	T	C	Mean
the city was	LSTM / greedy	“the first time of the first time in the united states, the first time in the united states...”	0	1	0	0	0.25
the city was	RNN / greedy	“a popular tourist attraction for the first time since the war. the first time, the storm was a tropical cyclone...”	2	2	2	1	1.75
the river flows	Self GloVe / greedy	“the missouri river river river river, the missouri river river, the missouri river river river...”	0	1	3	0	1.00
the river flows	Transformer / temperature 0.8	“through the city of omaha was a significant area of the city 's largest river avenue located near the confluence of kansas river...”	4	2	4	2	3.00
the river flows	Public GloVe / temperature 0.8	“is a first known as a ‘dry’. the fur is a metaphor used by a variety of staple birds...”	5	1	0	0	1.50

These examples illustrate that low mechanical repetition does not imply fluent or topically consistent text. The scores are from a single exploratory AI review (mostly based on opening excerpts), not independently validated ratings.

The full 288-row AI-assisted scoring record is available at results/ai_review/generation_ai_review_completed_288.csv. Original generations and decoding settings are available at results/generation/20261015_142451_559597/ (temperature sampling) and results/generation/20261015_143114_234189/ (greedy). Automatic per-generation statistics are in results/generation/quantitative_analysis/20261015_182026_757124/; prompt-group comparisons are in results/generation/group_analysis/.

Appendix C. Downstream Error Analysis

Table C1. Paired Transformer versus public GloVe prediction outcomes (2,000 test samples per dataset).

Dataset	Both correct	Transformer only correct	Public GloVe only correct	Both wrong	Net public gain
imdb	1143	203	297	357	+94
ag_news	1254	89	375	282	+286
dbpedia	1627	75	184	114	+109

Table C2. Most frequent off-diagonal confusion pairs for public GloVe.

Dataset	True label → predicted label	Cases
ag_news	Business → Sci/Tech	69
ag_news	Sci/Tech → Business	65
ag_news	Business → World	51
dbpedia	Animal → Plant	21
dbpedia	Plant → Animal	15
dbpedia	Artist → OfficeHolder	8

IMDb example A: the negative review containing “i have never fallen asleep whilst watching a movie before” was misclassified as positive by the random Transformer but correctly classified by public GloVe. IMDb example B: the positive review containing “badly acted and directed, poorly scripted. who cares?” was classified correctly by the Transformer but incorrectly as negative by public GloVe. These are selected illustrative cases, not an estimate of error-type prevalence.

Full class-level error rates, confusion pairs, and representative candidate records: results/part3/error_analysis/20261015_174111_594685/.

Appendix D. Exploratory Domain Similarity

Table D1. WikiText-2 training versus downstream training text: unigram statistics.

Dataset	JSD (bits) ↓	Shared token mass	Downstream tokens
imdb	0.33411	83.44%	934,364
ag_news	0.33703	83.94%	151,413
dbpedia	0.32011	81.45%	195,189

DBpedia exhibited the lowest unigram JSD but also the lowest shared token mass. Because DBpedia and WikiText-2 both draw on Wikipedia, potential overlaps were screened **before** constructing the downstream subsets. The audit extracted level-1 article headings from the WikiText-2 *training* split (629 heading occurrences; 620 unique titles), then compared them with the title field of all 560,000 DBpedia training and 70,000 test records. Titles were normalized using Unicode NFKC, case folding, WikiText punctuation-marker replacement, and whitespace/punctuation normalization. A strict comparison retained parenthetical disambiguation; a second, looser comparison removed a final parenthetical suffix and punctuation to catch ambiguous candidates.

DBpedia source split	Canonical-title matches	Additional loose-title candidates	Excluded candidates
Train (560,000 records)	60	121	**181**
Test (70,000 records)	6	23	**29**

All 210 candidate records were excluded before stratified downstream sampling, including ambiguous loose matches. The sampling script also skipped duplicate normalized title-plus-content texts across selected splits. **Title matching identifies possible overlap, not proven text duplication, and cannot rule out all near-duplicates.** See results/part3/dbpedia_overlap_v2/20261015_162003_934947/summary.json and matches.csv; the screening code is src/audit_dbpedia_overlap_v2.py and the downstream exclusion logic is in src/prepare_part3_dbpedia_numeric_fixed.py. These checks do not isolate domain effects because task labels, class counts, and document characteristics differ. Full similarity measurements: results/part3/domain_similarity/domain_similarity.csv.

File paths are relative to the project root. Files mentioned as separate attachments must be included in the final submission package if referenced here.

AI Use Statement

This assignment used only the GPT-6 model through a ChatGPT Edu account for experimental design discussions, code development, and report refinement.

During discussions of experimental design, GPT-6 demonstrated strong capabilities in proposing solutions but also a tendency to recommend particular approaches and make decisions prematurely. To address this, I explicitly prompted it to present multiple experimental alternatives, provide more conceptual explanations, and prioritize empirical testing over simply following its recommendations.

I made the vast majority of the final decisions and contributed numerous original ideas, including:

• Handling different hierarchical levels of text in WikiText-2;

• Preserving frequently occurring numerical tokens while replacing rare ones with <num>;

• Exploring multiple n-gram configurations through several rounds of hyperparameter tuning;

• Designing prompts for text generation evaluation;

• Selecting multiple downstream tasks with potentially different degrees of similarity to the upstream training corpus in Part III;

• Evaluating representations from all trained neural language models in Part III;

• Experimenting with multiple text generation strategies;

• Asking GPT-6 to evaluate generated texts using explicitly defined scoring criteria.

In programming, GPT-6 demonstrated substantially stronger coding capabilities than my own. However, it also showed a strong tendency toward defensive programming. As someone without practical software engineering experience, I chose to retain its highly conservative recommendations for multiple rounds of testing to reduce the risk of undetected implementation errors.

In experimental data processing and report preparation, GPT-6 was proficient in using Markdown and related tools. However, its initial interpretations of experimental results were often relatively superficial, and it was less effective at identifying the most meaningful findings and developing deeper analyses. Consequently, I took responsibility for interpreting the experimental results, identifying the key findings, and deciding the content and argument of each sentence. GPT-6 was then used for final language polishing and PDF generation.

GPT-6 also served as a single AI evaluator for an exploratory assessment of 288 generated continuations. The evaluation criteria, aggregation, and limitations are documented in the report and appendix.